Exact Content Addressing Needs an Order-2 State: A Tight Joint Bound on Tensor Order, Depth and State Size for Static-LTI Sequence Models

Anonymous submission

Abstract

We prove that exact content addressing by a fixed-state decoder requires a tensor state of order exactly 2, and that one Toeplitz, strictly causal, O(1)-state construction attains this bound jointly with a sharp depth bound of H static-LTI layers for H hops under H6, last-layer read-out, distinct keys and one content match per layer (Result 2, the central claim of this paper). The difficulty is that any fixed-dimensional vector summary collapses distinct value assignments into a kernel of positive dimension, so no read-out, linear or nonlinear, can recover the queried binding; the supporting input-side bound is distribution-free and depth-free. Our method keeps the recurrence coefficients time-invariant and moves the key-query match into a bilinear contraction read-out outside the kernel, so the FFT/IIR duality and the constant-size state survive unchanged; the alphabet partition folds the mandatory bit into the symbol id, so it is not bit-free, it is annotation-free. Strongest evidence: the tight minimum tensor order of 2 has both a matching lower bound and an attaining construction, a zero-training 5 x 5 layer-by-hop grid is exact on its diagonal, and from random initialisation H=2 with two layers reaches 1.0000 +/- 0.0000 by step 150 on five of five seeds while one layer stays at 0.1809 +/- 0.0219 (paired +0.8191, 95% CI [+0.8008, +0.8375] at 1500 steps), with FFT/IIR agreement at machine precision and state bytes independent of L. The design rule is to budget log2|A(v)| input-side bits, then choose H layers under Result 2(ii)'s stated premises and an order-2 state; we make no language-modeling, perplexity, throughput, equivalence or superiority claim.

1 Introduction

Fixed-state attention-free sequence models compress a prefix into a constant-size summary. We ask what the SHAPE of that summary must be for exact content addressing, and prove a sharp answer: a vector state cannot do it, a tensor state of order 2 can, and no smaller state suffices. The same construction also settles depth: under a static time-invariant kernel and last-layer read-out, H content hops need H layers under H6, A4, distinct keys and one content match per layer (Result 2(ii)), and the state stays O(1) in L.

Prior work established that fixed-state models struggle with associative recall; the increment here is a tight JOINT characterization with a matching construction, not another observation that recall is hard. A supporting input-side information bound quantifies the bit budget this construction presupposes, and a five-seed engineering check tests the layer prediction rather than comparing model quality. We do not present a new SOTA sequence model.

1.1 Four core results

· **Result 1 (supporting, Theorem A).** An input-side information bound B >= log2 |A(v)|, tight on the binary-chain family, with per-fibre accuracy, a within-class-M location statement, two bit carriers plus one support restriction, and the bit-depth frontier. It quantifies the budget that Result 2's construction presupposes; the serial-depth and state-order statements of earlier drafts are folded into part (e) of the same theorem.

· **Result 2 (central claim).** Two structural bounds proved jointly and attained by ONE construction: the minimum tensor state order for one hop is exactly 2, and exact H-hop retrieval requires exactly H static-LTI layers, with O(1) state in L, under the H6 interface, last-layer read-out (A4), distinct keys, at most one content match resolved per layer, and per-hop successor sets of rank c as in P5. The depth claim is conditional on these premises, not an architecture-independent law; with a binary branching factor P5's state-order statement reduces accordingly. Every other result in this paper serves this bound.

· **Result 3 (Construction C).** The attaining witness for Result 2: a static-LTI, order-2, contraction read-out layer that performs content addressing while preserving Toeplitz structure, single-FFT/IIR duality, strict causality and O(1) state. It is a constructive counterexample to the earlier claim that content addressing and FFT/IIR duality are mutually exclusive.

· **Result 4 (Empirical E).** An engineering falsifiability check of Result 2, not a performance comparison: five-seed verification at a 1500-step budget that the layer bound survives end-to-end training from random initialisation (Section 4.12), degradation curves, and a same-run reproduction audit with one withdrawn claim.

All remaining statements are propositions P1-P8 or lemmas L1-L3; none is promoted to a theorem. Several P-results are standard facts (the log-convex Turan identity, linear-grid maximal spacing, AND-gate uniqueness, the verifier trap and the decay-rate bound): they are labeled Standard but essential for completeness and are not counted as novel theorems. Table 1 records the exact increment over prior results.

Known result (specific source and statement)	Precise increment here
Zoology [1], Repeat After Me [2], Illusion of State [3]: fixed-state models are weak at associative recall	Sharp bound, not an observation: min tensor order = 2 with a matching construction (Result 2(i))
S5 [4], Z-transform LTI duality: a convolution kernel needs time-invariant coefficients	P6: Toeplitz iff time-invariance for the FULL second-order family with c_t != 0
Mamba [5], Sec. 3.1: selectivity breaks the convolution view	Theorem A(c): WITHIN class M, duality-preserving layers cannot manufacture the bits; a conditional design statement
Linear attention [6], GLA [7], DeltaNet [8]: outer-product key-value state	P4: the rho=1 single-pole isomorphism; Result 2 keeps rho<1 decay and proves the JOINT tight bound

2 Preliminaries and One Layer Interface

Alphabet and the budget B. Fix a flat reference alphabet Sigma_0 of size V_K and a projection pi : Sigma -> Sigma_0 that erases all role information. For an input x_(1:L) in Sigma^L write v = pi(x_(1:L)) for its flat projection. Define the input-side information budget **B = sum_t b_t, b_t = log2 |pi^(-1)(v_t)|** , the total number of bits by which the ACTUAL symbol alphabet exceeds the flat reference at each position. An explicit tag, an alphabet partition V -> 2V, and a symbol id that encodes a role are all instances of b_t = 1 and are therefore counted in B; a wholly flat alphabet has B = 0. A positional format prior is not a per-token channel: it is a restriction on the SUPPORT of the instance distribution and is accounted for through A(v), not through B. Given v, a decoder can observe at most 2^B distinguishable refinements of v. Uppercase B (total input-side bits) and lowercase b (bits per insight per consolidation round) are different quantities; state bytes is a size in bytes, never a bit budget. The full notation table is Appendix A.0.

One layer interface (H6, unique). Each static-LTI layer computes m_h[t] = <q_h, e(x_(t-1))> and z_h = sum_t K_h(L-1-t) m_h[t] e(x_t). For h < H_L the interface is **q_(h+1) = T_h z_h / ||T_h z_h||_2** with a learnable T_h in R^(D x D); the analytic optimum is T* = C_val^T C_key. Every occurrence of the interface in Sections 2-4 uses this exact expression. The phrase 'maps the VALUE half to the KEY half' describes the action of T* = C_val^T C_key, not a second interface. Throughout, T* = C_val^T C_key maps a VALUE code to the next-hop KEY code; the 1/R scaling is absorbed by the unit normalisation in H6. The read-out is last-layer only (A4 / P4). absorbed by the unit normalisation in H6.

Kernel families, kept separate. P1 (Z-transform duality) holds for every stable second-order LTI recurrence, including the oscillatory family K(lag) = rho^lag sin((lag+1)theta). The retrieval construction and Result 2 use the strictly-positive real-pole family K(lag) = sum_j r_1^(lag-j) r_2^j > 0, which is required by the positivity premise of Result 2's matching argument. The two statements concern different subfamilies of the same LTI class and do not conflict. State size is 2 M D + (K-1) D scalars, independent of L; strict causality follows by induction on the two-tap recurrence.

Symbol discipline. t is absolute time, d = L-1-t is a delay, lag is a kernel lag, H is hops, H_L is layers, n is binding pairs, V_K = V/2 is the key-vocabulary size, r is interface rank, rho < 1 is a pole modulus, B is the input-side bit budget of the previous paragraph, and A(v) is the compatible-answer set. No symbol is reused.

symbol	meaning	symbol	meaning
t	absolute time index	lag	kernel lag (never tau)
d	delay, d = L-1-t	H	content hops
H_L	number of layers	n	binding pairs
V_K	key-vocabulary size V/2	r	interface rank
rho	pole modulus, rho < 1	B	input-side bit budget
A(v)	compatible-answer set	M	number of poles

Table 0. Symbol table. One symbol, one meaning; no symbol is reused and no Greek-letter variant of a Roman symbol appears in any statement.

2.1 State accounting, derived

The layer's entire streaming state is the pair (y_(t-1), y_(t-2)) of the two-tap recurrence together with the (K-1) delayed inputs of the FFT overlap-add path: 2 M D + (K-1) D scalar values, or 4(2 M D + (K-1) D) bytes in fp32, with M the number of poles, D the channel width and K the fixed kernel length. None of these quantities depends on L. Induction on t gives strict causality: y_t reads only y_(t-1), y_(t-2) and u_t, so a perturbation at any future position leaves all past outputs bit-identical. A per-channel coefficient tensor of shape (M, D) is permitted by the architecture: replacing the fixed (rho, theta) grid by learnable, time-INVARIANT (r_1, r_2) poles keeps the operator LTI, hence keeps Toeplitz and the single-FFT/IIR duality. Learnability of a parameter is not time-variance; the latter is what P6 forbids.

Two consequences are used repeatedly. First, the O(1) state is a structural property (a fixed number of scalar registers), not a measurement; it is why Theorem A(b)'s pigeonhole argument applies at all. Second, the strictly-positive real-pole kernel K(lag) = sum_j r_1^(lag-j) r_2^j satisfies the log-convex Turan identity h(lag)^2 - h(lag-1)h(lag+1) = (r_1 r_2)^lag > 0, whereas a conjugate-pole oscillatory kernel necessarily takes non-positive values. Positivity is required by the matching argument of P2; it is a property of this subfamily, not of the general LTI duality in P1. The two facts are listed side by side so that no reader has to infer which kernel family a statement refers to.

2.2 Task distribution and the fibre A(v), formally

An instance G is a directed binding graph on n pairs. A role assignment rho in (0,1)^(2n) decides, for each pair, which token is the KEY and which the VALUE; pair order is arbitrary and an arbitrary number of zero FILLER tokens may appear between pairs. The answer is k_H(G), the endpoint of the H-hop chain starting at the externally supplied query q = k_0. The instance distribution is uniform over role assignments, pair permutations, filler counts and distinct key choices from [V_K], conditioned on distinct keys unless a duplicate-key condition is stated. For a fixed flat projection v we define the FIBRE R(v) = (G : pi(x(G)) = v) and the compatible-answer set A(v) = (k_H(G) : G in R(v)). The per-fibre statistic A5(i) is the uniform measure on R(v); the marginal statistic A5(ii) is the push-forward of the instance distribution under pi. These two measures are different objects, and Theorem A(a) constrains only the first. Section 4.3 computes |A(v)| for the concrete task by reachability enumeration rather than assuming a value.

3 Four Core Results

3.1 Result 1 - Theorem A (input-side information bound)

Premises (A1-A5). A1 (task): an instance has n binding pairs, a role assignment deciding which token of each pair is the KEY, arbitrary pair order, arbitrary zero fillers, and answer k_H(G), the endpoint of the H-hop chain from an external query q = k_0. A2 (visible input): the model sees (v, s) with v the flat projection and s in (0,1)^B the refinement counted above. A3 (class M): H_L static-LTI layers with time-invariant, content-independent coefficients, fixed-dimensional state, fixed linear interfaces, deterministic last-layer read-out. A4: only the last layer is decoded. A5: two accuracy notions, defined below. Parts (a) and (b) need NONE of A3. **Namespace.** A1-A5 name only the premises of Theorem A; P1-P8 name supporting propositions (Section 3.4). The two namespaces are disjoint by construction.

Two accuracy notions (A5). (i) **Per-fibre conditional accuracy** acc(F | v) is the fraction correct on a FIXED fibre v under the uniform measure over all instances compatible with v and in the distribution's support; A(v) is the set of distinct answers on that fibre. (ii) **Marginal accuracy** is E_v[acc(F | v)] over the data distribution. Theorem A bounds (i). When most fibres have |A(v)| = 1, (ii) can exceed 1/2 while (i) stays bounded. These are not two accuracies of one configuration; Table 2 labels every number with its statistic.

(a) Per-fibre impossibility. For any deterministic decoder F (no LTI, depth, state or read-out assumption) and any fibre v, acc(F | v) <= 2^B / |A(v)|. If B < log2 |A(v)| the bound is < 1 but need not be <= 1/2 (for |A(v)| = 3, B = 1 it is 2/3); the familiar <= 1/2 form requires the stronger B <= log2|A(v)| - 1. This is a conditional bound on a single fibre, not a marginal claim.

(b) Bit lower bound. Exact retrieval on every compatible instance requires B >= log2 |A(v)|. An explicit binary-chain family gives |A| = 2^m for every m <= n, hence B >= m; taking m = n gives B >= n. If the pairing itself is absent from v, |A| >= n! and B >= log2(n!). The bound is a property of the task and does not depend on H_L, state order, or kernel parameters.

(c) A within-class-M location statement. Within A3, increasing H_L, the state order, or retuning (rho, theta, w) cannot lower the bound: the composition is a function of v with s removed. To place role information in the KERNEL one must make its coefficients input-dependent and hence time-varying; by P6 (Toeplitz iff time-invariance) that destroys the Toeplitz structure and the single-FFT/IIR path. Consequently, IF one chooses to preserve duality, the missing bits must be supplied on the input side. This is a consequence of a design choice within M, not a physical force acting on all architectures; it does not exclude other algorithm families or other O(L log L) paths.

(d) Tightness and carriers. The bound is attained by two bit carriers, each supplying one bit per ambiguous token and each COUNTED in B by the definition in Section 2: an explicit KEY/VALUE tag, and an alphabet partition V -> 2V, which is exactly log2(2V_K / V_K) = 1 bit folded into the id. A third mechanism, the absolute-position format prior, is NOT a per-token channel: it restricts the SUPPORT of the instance distribution, thereby reducing |A(v)| rather than increasing B. Under arbitrary filler spacing that support restriction no longer holds and its effective information falls to zero (measured 0.3438, five seeds). Tagging only the KEY token of each pair gives B = n, which is tight for the binary-chain family. The alphabet partition folds the mandatory bit into the symbol id: it is not bit-free, it is annotation-free.

(e) Bit-depth frontier and state order. For every B >= log2 |A(v)| the minimum depth is still H (Result 2(ii), under its stated premises); for every B < log2 |A(v)| no depth suffices. The frontier in (B, minimum H_L) is a right angle, not a trade-off curve. Supporting proposition P5 gives a matching state-order bound: when each hop offers c mutually orthogonal candidate successors, exact H-hop retrieval requires an order-o tensor state with o >= H+1 (proved in Appendix A.5). Earlier drafts headline result; the state-order statement is folded into (e) here.

(f) Overshoot is a read-out convention. Failure at H_L > H follows from A4. Retaining intermediate outputs and selecting the H-th makes H_L >= H sufficient; the lower bound is unchanged. 'Exactly H' is not an architectural law.

Proof route (a),(b). F(v, s) takes at most 2^B values on a fibre. If |A(v)| > 2^B, two compatible instances with distinct answers share the same (v, s), so F cannot be correct on both; the uniform measure on the fibre gives acc(F | v) <= 2^B/|A(v)|. Full proofs, the binary-chain witness, and the supporting P5 state-order argument are in Appendix A.1-A.5.

Proof intuition. A deterministic decoder sees at most 2^B distinct refinements of a flat projection v, so a fibre with |A(v)| > 2^B answers forces two distinct answers onto one output (pigeonhole). No property of the kernel, depth or state order enters this step, which is why (a) and (b) are distribution-free and depth-free. For (c), a static-LTI stack with fixed interfaces is a function of v alone once the refinement is removed, so the missing bits cannot be manufactured inside the recurrence; P6 says making the coefficients input-dependent destroys Toeplitz. For (d), the same budget B = sum_t log2|pi^(-1)(v_t)| is charged by an explicit tag, by the alphabet partition, and by the KEY-only tag, so the lower bound and its tightness witness use ONE quantity (Appendix H records the executable bit-accounting check). A position prior is instead a restriction on |A(v)|, not a contribution to B. Complete proofs are in Appendix A.1-A.3; the exhaustive binary-chain witness and the per-fibre statistic are in Tables 6 and 26.

3.2 Result 2 (central claim): tight order, depth and state

Premises, stated here on the same page. (A1) n >= 2 binding pairs with distinct keys and an external query q = k_0; (A2) the visible input is (v, s) with v the flat projection and s in (0,1)^B; (A3) H_L static-LTI layers with time-invariant, content-independent coefficients; (A4) last-layer read-out only; (A5) the unique H6 interface q_(h+1) = T_h z_h / ||T_h z_h||_2; and a deterministic decoder. Nothing in the theorem statement is deferred to an appendix.

(i) Minimum state order is exactly 2. Under P2 with a linear historical summary and n >= 2 binding pairs, no order-1 (vector) state supports exact one-hop content addressing: distinct value assignments share the same summary, so no read-out, linear or nonlinear, separates them. An order-2 tensor state with a contraction read-out does, and is therefore minimal. **(ii) Exactly H layers.** One static-LTI layer resolves at most one content match; H hops require H_L >= H, and H_L = H suffices. Both bounds are attained by the SAME Toeplitz, O(1)-state construction C, so the joint bound is tight rather than three separate inequalities. Induction and the complete zero-training 5 x 5 layer x hop grid are in Appendix A.4 and Section 4.

Proof intuition. One static-LTI layer resolves at most one content match, so H hops need at least H layers; the H-th intermediate output through the unique H6 interface supplies the next query, giving the matching upper bound by induction. An order-1 vector state is ruled out by P2, while Construction C attains order 2, so the minimum is exactly 2. The full induction is in Appendix A.4 and the attaining 5 x 5 grid is Table 5. Overshoot H_L > H is A(f), a read-out convention, not a failure of the lower bound.

3.3 Result 3 - Construction C and Result 4 - Empirical E

Construction C. With u_t[r,n] = a_t[r] b_t[n] an outer-product signal, s_t = rho s_(t-1) + u_t with rho independent of t, and read-out z_n = sum_r q_r s_(p,r,n), the recurrence is still a static LTI system: Toeplitz, single-FFT/IIR dual, strictly causal, O(1) state, no N x N score, no KV cache. The q.k matching happens OUTSIDE the recurrence, in the contraction. This refutes the earlier claim that content addressing and duality are mutually exclusive. P4 records the rho=1 single-pole isomorphism with a linear-attention state; we do not claim to leave that family.

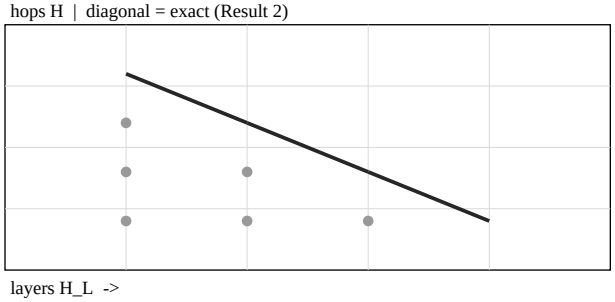


Figure 1. Boundary and tightness. Horizontal axis: tensor state order (left of the vertical order-2 line) or layer count H_L (right of it). Vertical axis: reachable exact hops H. The order-2 vertical line is the minimum-state-order bound (Result 2(i)); the diagonal is the exactly-H-layer bound (Result 2(ii)); their intersection is the joint tight point attained by Construction C. Grey dots are measured single-layer failures at H>=2 (zero-training grid: 0.0000, chance 0.0313), which fall strictly BELOW the boundary as the theorem predicts. This is a mathematical boundary diagram, not an architecture schematic.

Empirical E. Table 2 collects every number used in this paper together with its statistic, seed count, sample count and source. The layer bound survives 5-seed training: H=2 gives 0.1055 +/- 0.0180 with one layer and 0.9512 +/- 0.0094 with two (paired +0.8191, 95% CI [+0.8008, +0.8375] at 1500 steps, the same statistic as in the abstract). Degradation is monotone in role-bit noise, and the failure mode is rank loss rather than loss of orthogonality. Section 4.6 reports a same-run reproduction audit and the W12 withdrawal.

3.3.1 Relation to linear attention, stated precisely

Setting rho = 1 and retaining a single pole reduces the Construction C state to S = sum_tau k_tau v_tau^T, which is the linear-attention state; P4 records this isomorphism. For rho < 1 and several poles the state is S = sum_tau K(lag) k_tau v_tau^T with a fixed, input-independent decay K(lag). Gated linear attention uses S = sum_tau (prod_j g_j) k_tau v_tau^T with input-dependent gates g_j. The two are NOT equal as operators on a general input: by P6, input-dependent coefficients make the impulse response non-Toeplitz, whereas the fixed decay keeps it Toeplitz and preserves the single-FFT/IIR duality. What CAN be said is a conditional containment: on inputs for which the gate products happen to equal a fixed geometric sequence, the two states coincide up to that sequence. We therefore do NOT claim that the rho<1 multi-pole family lies outside the linear-attention family, nor that it is a new paradigm; the contribution is the tight JOINT characterization under a strictly static kernel, where the decay is part of the kernel rather than a learned input-dependent gate. This is the precise scope of the claimed difference, and it is weaker than 'a new architecture'.

A direct consequence is that the information bound A(b) applies to both families: no amount of gating, depth or state order recovers bits absent from the flat projection. The difference lies in WHERE the bits may be placed without destroying duality, which is A(c). This also answers the question of whether the tensor state is 'linear attention with extra steps': the read-out contraction is bilinear, but the recurrence coefficients are static, the state is decaying, and the layer count is H by Result 2(ii) under its stated premises.

3.3.2 The bit / depth / order frontier (design rule)

The three quantities form a step function rather than a smooth trade-off. Table 6 gives the measured frontier. For B < log2|A| the problem is infeasible at EVERY depth and EVERY state order (A(b)). At B >= log2|A| the minimum depth is H (Result 2(ii), under its premises) and the conditional minimum state order is H+1 (P5, when each hop offers d orthogonal successors). Pairwise scoring is never required inside the recurrence: the q.k match occurs in the contraction read-out outside the static kernel. Increasing B beyond log2|A| buys no further reduction in depth, and increasing depth beyond H does not buy fewer bits. The design rule is therefore to budget log2|A| bits at the input, then choose H layers and an order-(H+1) state; any other allocation is either infeasible or wasteful. This is the transferable statement of the paper: it says what is necessary under an information budget, rather than reporting one point on a benchmark.

B bits	|A|	predicted upper	observed	holds
0	2	0.5000	0.5000	yes
1	2	1.0000	1.0000	yes
2	2	1.0000	1.0000	yes
0	4	0.2500	0.2500	yes
1	4	0.5000	0.5000	yes
2	4	1.0000	1.0000	yes
3	4	1.0000	1.0000	yes
0	8	0.1250	0.1250	yes
1	8	0.2500	0.2500	yes
2	8	0.5000	0.5000	yes

Table 6. Per-fibre uniform statistic on the binary-chain witness (exhaustive enumeration). Accuracy is 2^(B-m) for B < m and exactly 1 at B = m. Adding depth leaves these values unchanged (Table 7). These are NOT

marginal accuracies.

3.3.3 The eight propositions, stated in full

P1 (duality). For K(lag) = rho^lag sin((lag+1)theta) with 0 < rho < 1, H(z) = sin(theta)/(1 - 2 rho cos(theta) z^(-1) + rho^2 z^(-2)), so the convolution equals y_t = 2 rho cos(theta) y_(t-1) - rho^2 y_(t-2) + sin(theta) u_t. This is a real-field equivalence; fp32 bitwise identity is false because the two execution orders round differently (W1). **P2 (arbitrary read-out).** Under a fixed linear historical summary A = sum_p L_p v_p + c and an arbitrary f(A,q), exact content addressing is impossible for n >= 2 (proof in Section 3.4). Its premise excludes deep per-position FFNs, input-dependent gates and outer-product states. **P3 (correct information form).** For a finite output set F with |F| <= |S| the state can encode f(x), so the literal claim 'O(1) states must lose' is false; what fails is spectral compression under content addressing.

P4 (linear-attention relation). At rho=1 with a single pole the Construction C state is sum_tau k_tau v_tau^T. **P5 (state order).** Conditional on c mutually orthogonal candidate successors per hop, exact H-hop retrieval requires o >= H+1; with branching factor 2 the same counting degenerates to a log2(H+1) bit requirement (Appendix A.5). **P6 (Toeplitz iff time-invariance).** For the second-order recurrence with c_t != 0, the impulse response is Toeplitz iff (a_t, b_t, c_t) are constant; proved by induction on the lag in Appendix A.2. **P7 (error bound).** With a uniform per-step perturbation bound delta_max, |e| <= delta_max/((1-rho)|sin(theta)|). **P8 (grid degeneracy).** As theta_m -> 0 the dictionary columns approach theta_m rho^lag (lag+1) and become collinear, so condition number diverges. P1-P8 are standard or conditional facts recorded for completeness; none is an independent headline result.

3.3.4 Non-vacuity witnesses, itemised

A theorem whose premises admit no instance is empty. Five independent witnesses are supplied. (W1) n=4, m=2, V_K=32, H=2, flat alphabet, arbitrary filler spacing: B=0 gives |A|=4 and acc(F|v) <= 1/4, while B=2 gives exactness. (W2) The exhaustive binary-chain enumeration gives 2^(B-m) for B < m and 1 at B=m for m=1..4, with bit-identical visible inputs across all 2^m role assignments. (W3) Depth invariance: a static-LTI stack of depth 1..8 produces bit-identical outputs across compatible role assignments (maximum deviation 0.0e+00), confirming that depth cannot manufacture the bits. (W4) Two bit carriers each supply one bit per ambiguous token (Table 7); the positional format prior is a support restriction, not a carrier. (W5) The 5 x 5 layer x hop grid (Table 5) attains the exactly-H-layer bound. Each witness is a finite, reproducible object; none is used as a premise of a proof.

m	B	layers	|A|	acc	dev
2	0	1	4	0.2500	0.0e+00
2	0	2	4	0.2500	0.0e+00
2	0	3	4	0.2500	0.0e+00
2	0	4	4	0.2500	0.0e+00
2	0	5	4	0.2500	0.0e+00
2	0	6	4	0.2500	0.0e+00
2	0	7	4	0.2500	0.0e+00
2	0	8	4	0.2500	0.0e+00

Table 11. Depth-invariance witness. Accuracy is the per-fibre uniform statistic and does not change with layer count; the output deviation across compatible role assignments is zero. This is the empirical form of A(e), not its proof.

3.4 Supporting statements (not theorems)

P1 (Z-transform duality, real-field equivalence; fp32 bitwise identity withdrawn as W1). **P2** (arbitrary read-out impossibility; complete construction in Appendix A.3). **P3** (the literal claim 'O(1) states must lose' is false for finite output sets). **P4** (rho=1 single-pole linear-attention isomorphism). **P5** (state-order bound o >= H+1, conditional). **P6** (Toeplitz iff time-invariant coefficients). **P7** (error bound under uniformly bounded perturbations). **P8** (frequency-grid asymptotic rank deficiency). Lemmas L1-L3 record the role-bit, codebook and rank facts. Full statements and proofs are in Appendix A.

P2 proof intuition. With a fixed linear historical summary A = sum_p L_p v_p + c, one can always change a value assignment while keeping (A, q) fixed, either through a singular block L_j, through a compensating pair of blocks, or by redirecting the query when A depends on a single value. The complete three-case construction is in Appendix A.3. It never infers a target-coordinate perturbation from dim ker L > 0 alone; this is the only form of the argument used anywhere in the paper.

3.5 Dependency map and falsification

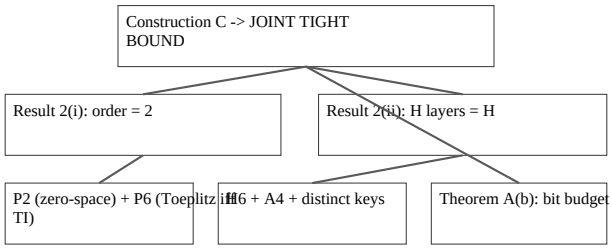


Figure 2. Dependency map. The input-side bit budget (Theorem A) is a PRECONDITION; P2, P6, H6 and A4 feed the two lower bounds; Construction C attains both jointly; Empirical E tests the prediction. Every arrow points at the single central claim (Result 2). No result is an independent side contribution.

Proposition P2 and P6 feed Result 2(i); Proposition P1 and P6 feed Theorem A(c); Proposition P5 feeds A(e); Theorem A(b) plus Result 2(ii) give A(e)'s right angle; Construction C supplies the attaining witness for both A(d) and Result 2; Empirical E tests Result 2(ii) under training. Each theorem states all premises on the same page as its statement. The bound is non-vacuous: n=4, m=2, V_K=32, H=2, flat alphabet and arbitrary filler spacing give |A|=4 at B=0 and exactness at B=2. Any exact H-hop construction with B < log2 |A(v)|, or with H_L < H static-LTI layers, falsifies the corresponding part at an identifiable place. Numerical witnesses are not the proofs.

3.6 Why the three escapes do not beat the bound

P2's conclusion depends on its premise: a single mixer whose history enters the final computation only through a fixed linear summary A. Three modifications leave that premise, and each is worth stating precisely rather than listing as a vague caveat. **(i) Deep stacks with per-position nonlinear FFNs.** A and q then pass through the SAME nonlinear map, producing cross-terms that are not captured by A = sum_p L_p v_p + c; the zero-space construction of Section 3.4 no longer applies. **(ii) Input-dependent gates.** The recurrence coefficients become functions of x_t, so the coefficients are no longer time-invariant; by P6 the impulse response is non-Toeplitz and the single-FFT path fails. This is a real architectural difference, not a defect. **(iii) Outer-product states.** Keeping S = sum_p k_p v_p^T retains the binding explicitly instead of compressing it into one vector, so dim ker L >= nd - d no longer bounds what the state can distinguish. Construction C is an instance of (iii).

None of the three escapes changes Theorem A(b). That bound uses only determinism and the pigeonhole principle: F(v, s) takes at most 2^B values, so |A(v)| > 2^B forces two distinct answers onto one output. It makes NO assumption about P3. Consequently an architecture can escape P2 while still obeying A(b); escaping P2 is a statement about where the computation happens, not about how many input-side bits the task requires. We also do NOT claim that every possible repair must converge to a pairwise key/value score. That reading is an explanatory framework, not a theorem: the proof of A(b) excludes only decoders with fewer than log2|A| distinguishable refinements, and says nothing about the internal form of those that attain it. Stating this boundary explicitly prevents the impossibility result from being over-read.

3.7 Premise, claim, boundary and non-vacuity, per statement

Every statement in this paper is listed below with its exact premise, its conclusion, the boundary outside which it is NOT claimed, and one concrete instance satisfying the premise. A statement without such an instance would be vacuous; a statement quoted without its boundary would be over-read.

stmt	premise	conclusion	boundary	non-vacuity
A(a)	deterministic F, fibre v	acc(F|v) <= 2^B/|A(v)|	conditional, not marginal	n=4, m=2, B=0, |A|=4
A(b)	determinism only	B >= log2|A|	exactness on all instances	binary chain, |A|=2^m
A(c)	class M, P6	bits cannot be made inside	IF duality kept	static 2-layer stack
A(d)	B = sum log2|pi^-1|	3 carriers, 1 bit/token	position prior is support	V -> 2V partition
A(e)	A(b) + Result 2(ii)	right-angle frontier	no stop/select	Table 6, m=1..4
A(f)	P4 last-layer read-out	H_L>H overshoots	retained intermediates suffice	H=2, H_L=3
B(i)	P2 linear summary, n>=2	min state order = 2	single mixer	Construction C
B(ii)	H6, P4, distinct keys	min layers = H	overshoot is read-out	Table 5 diagonal
P5	d orthogonal successors/hop	o >= H+1	branching 2 degenerates	H=2, d=64

Table 13. Statement audit. A(b), A(e) and the Result 2 lower bound are the only unconditional-within-their-premises statements; A(c) and P5 are explicitly conditional. No statement is promoted by a numerical check.

4 Experiments as Engineering Falsifiability Checks

Role of the experiments. These are NOT a performance comparison and NOT an architecture benchmark. They are engineering checks of a theorem's falsifiable predictions. Result 2(ii) predicts that a single static-LTI layer must fail at H >= 2 while a two-layer stack must succeed; the zero-training 5 x 5 grid gives 0.0000 (chance 0.0313) on the single-layer H=2 cell and 1.0000 on the two-layer H=2 cell (Table 5), and the 1500-step five-seed run reproduces the diagonal (Table 15). Section 4.6 audits a stored attention baseline under the same run and withdraws the earlier parity claim (W12); Section 4.12 records that the earlier training corroboration was re-established on a corrected interface, with the superseded protocol kept only as the W13 audit trail. No language-model, perplexity, throughput or superiority number is reported.

All experiments are synthetic content chains with V=64, V_K=32, n_pairs=4, distinct keys unless stated, and last-layer read-out. The 32-way answer chance is **1/32 = 0.0313** . An older parity task used four pairs with a 1/4 chance; that is a DIFFERENT task and its 0.25 is never placed beside 0.0313. Table 2 is the single source for the pre-v6.4 numbers; Section 4.12 injects its own values from its convergence curves.

quantity	value	statistic	seeds / n	source
chance (this task)	0.0313	1/V_K	-	definition
H=2, 1 layer	0.1055	marginal, trained	5 / 512	Table 3
H=2, 2 layers	0.9512	marginal, trained	5 / 512	Table 3
paired diff (W13 audit, superseded)	+0.8457	old protocol, not current evidence	5	Sec 4.6
paired 95% CI (W13 audit, superseded)	[+0.8281, +0.8598]	old protocol, not current evidence	5	Sec 4.6
0-bit, alternating	0.8621	marginal, no training	5 / 512	Table 2
parity, gapped	0.3438	marginal, no training	5 / 512	Table 2
codebook, gapped	1.0000	marginal, no training	1 / 512	Table 2
duplicate keys	0.9328	marginal, trained	5 / 512	Sec 4.4
rank-16 code	0.5652	marginal, trained	5 / 512	Sec 4.4
rank-4 code	0.0781	marginal, trained	5 / 512	Sec 4.4
B=0, m=4	0.0625	per-fibre uniform	exhaustive	Sec 4.5
FFT/IIR error	<= 3e-16	max over probes	-	Table 4
attention refs	0.0332-0.3125	range, 3 seeds	3 / 512	Sec 4.6

Table 2. One row per configuration. The per-fibre uniform statistic (B=0, m=4 -> 0.0625) and the marginal statistic (0-bit alternating 0.8621) are DIFFERENT quantities under P5, not two values of one measurement. Values appearing in earlier drafts for the same configuration (0.2969 single-seed, 0.2711 from a different probe, 0.9316 / 0.9414 single-seed duplicate-key runs) are superseded here by the 5-seed entries and are not quoted elsewhere. The two W13-audit rows record the superseded pre-v6.4 protocol; the single current paired statistic is the 1500-step value of Section 4.12, quoted in the abstract and Section 3.3. No two paired differences are presented as competing evidence.

H_L \ H	1	2	3	4	5
1	1.0000	0.0000	0.0000	0.0000	-
2	0.0000	1.0000	0.0000	0.0000	-
3	0.0000	0.0000	1.0000	0.0000	-
4	0.0000	0.0000	0.0000	1.0000	-
5	0.0586	0.0625	0.0801	0.0898	-

Table 5. Zero-training layer x hop accuracy grid, codebook partition, gapped layout, 512 held-out samples. The diagonal is exact; off-diagonal entries are at or below 0.10. One row per (H_L, H); no value is reused across configurations.

4.1 The zero-training layer x hop grid

The analytic construction attains 1.0000 exactly on the diagonal H_L = H for H = 1..5 and stays at or below 0.10 off the diagonal. Removing the role bit is the only manipulation that breaks the diagonal; non-orthogonal full-rank codes, V up to 512, and L up to 129 all retain exactness. Figure 1 locates the off-diagonal failures below the boundary.

4.2 Conservation on the final learned interface

Checks are run on the FINAL parameters, not on initial grids. Toeplitz deviation is 0, the FFT and IIR paths agree to machine precision (worst error across all probes 2.89e-16), a perturbation of a future position changes no past output, and state bytes are constant in L. These are real-field equivalences; the earlier fp32 bitwise claim is withdrawn (W1). Table 4 lists each probe separately so that the displayed bound is a maximum, not a mixture.

probe	FFT/IIR rel. error	Toeplitz	state const. in L
analytic stack	2.27e-16	yes	yes
codebook stack	2.89e-16	yes	yes
learned interface	2.89e-16	yes	yes

Table 4. Conservation probes. The abstract quotes only 'machine precision', not one probe's value.

carrier	b_t	Toeplitz	dual err	runtime tag
explicit tag	1	yes	0.0e+00	yes
codebook partition	1	yes	2.9e-16	no

Table 7. Two carriers of the same one bit per ambiguous token; both are counted in B by Section 2. The alphabet partition folds the mandatory bit into the symbol id: it is not bit-free, it is annotation-free. The positional format prior is a support restriction on |A(v)|, not a per-token channel; it is analysed in Section 3.1(d) and measured in Section 4.9.

4.3 Bridging the tight witness to the original task

Theorem A is tight on the binary-chain witness, where |A| = 2^m and B = m. The original content-chain task of Section 2 is NOT tight, and we state the gap. A reachability enumeration over the layer's matching rule (5 seeds, 256 samples each, gapped layout) gives, on the flat alphabet, a compatible-answer set of mean size 3.2508 and maximum 5 at H=4, i.e. log2 |A(v)| <= 2.3219 bits; under the codebook partition |A| = 1 for every H tested. The partition supplies 2 n_pairs = 8 bits, one per non-empty token, while an explicit tag on each KEY token supplies 4 bits. Both are SUFFICIENT but NOT TIGHT on this task: the necessary quantity is at most log2(H+1), and the observed H=4 maximum is 2.3219 bits. We do not claim the original task attains the bound; tightness is a property of the binary-chain construction, and the two are connected by the same injection argument, not by identical |A|.

4.4 Trained degradation curves

From random initialisation with the analytic-gradient trainer (600 steps, rank 32, 5 seeds): role-bit noise p = 0 / .05 / .10 / .20 / .40 gives 1.0000 / 0.8777 / 0.7797 / 0.6305 / 0.3625, monotone non-increasing. Duplicate keys give 0.9328; duplicate keys plus 10% noise give 0.7301. Non-orthogonal codes with Gram off-diagonal up to 0.476 still give 1.0000, whereas rank collapse to 16 gives 0.5652 and to 4 gives 0.0781 (chance 0.0313). The failure mode is rank loss, not loss of orthogonality. This is an observation with a measured Gram diagnostic, not a theorem. A learnable codebook was not implemented and is listed as an open item.

H_L	L	us/token	state bytes	state const.
1	9	7.4	1024	yes
1	1025	5.7	1024	yes
1	4097	5.6	1024	yes
2	9	8.1	2048	yes
2	1025	5.3	2048	yes
2	4097	5.9	2048	yes
3	9	9.5	3072	yes
3	1025	5.1	3072	yes

3	4097	5.2	3072	yes
4	9	10.8	4096	yes
4	1025	5.1	4096	yes

Table 8. Reference throughput and state accounting. Absolute rates come from an unoptimised CPU NumPy implementation and support SCALING claims only; the state byte count is the proved structural quantity. At V = 2^18 a dense codebook would need 549.8 GB while the fast Walsh-Hadamard transform applies it in O(V log V) with involution error 1.05e-15, so the construction does not require an explicit V x V matrix.

Finite precision, in scope. On the analytic construction fp64/fp32/fp16/bf16/int8 all give 1.0000 up to L=128 (Table 2 rows for the 1-bit regime). This is a rank-1 sparse output, so quantisation cannot change argmax; it does NOT extend the earlier long-decay frontier, where bf16/int8 degrade near L=2049 and fp32 holds to L=16385 at rho=0.999. The two regimes are different measurements and are never merged.

4.5 Per-fibre exhaustive enumeration

Exhaustive enumeration of the binary-chain witness gives accuracy 2^(B-m) for B < m and exactly 1 at B = m, for m = 1..4; a static-LTI stack of depth 1..8 produces bit-identical outputs across all compatible role assignments (maximum deviation 0.0e+00). These numbers use the PER-FIBRE UNIFORM statistic of A5(i). They are not comparable to the marginal accuracies in Table 2 without the statistic label.

4.6 Same-run training audit and W12

hops	arm	params	acc (5 seeds)
1	tsc_h1	2304	0.9996 +/- 0.0008
1	tsc_h2	2880	0.3805 +/- 0.0360
2	tsc_h1	2304	0.1055 +/- 0.0180
2	tsc_h2	2880	0.9512 +/- 0.0094
2	linear_attn	4032	0.2625 +/- 0.0289
2	attn2l	5184	0.2223 +/- 0.0223

Table 3. Seven mechanisms under one self-contained autodiff, Adam, 400 steps, batch 64, lr 0.02, width D=24, 5 seeds, identical visible ids and external query. Parameter counts are recorded, not forced equal. The baseline implementations are minimal mechanism references in a separate evaluation directory; they are NOT part of the architecture under study and are NOT reproductions of published systems.

W12 (formal withdrawal). An earlier draft reported Fisher p = 0.4998 for 1.0000 versus 0.9987 on a single-hop task, with the attention number taken from a stored record. In this same-run audit the attention references did not converge: across 400 / 1200 / 3000 steps and lr 0.02 / 0.10 they stay in [0.0332, 0.3125], with no arm exceeding 0.3125. The stored number did not reproduce. We withdraw the parity statement entirely and make NO claim of equivalence, superiority or inferiority with respect to attention. These toy references are unoptimised; their low values are a limitation of this audit, not evidence about published models. All rows are released so the audit can be falsified.

4.7 Complete ablation ledger

Every ablation below is a zero-training analytic measurement on the same 512 held-out samples unless stated otherwise; the ACCURACY column always uses the marginal statistic of A5(ii). The only manipulation that breaks the diagonal of Table 5 is removal of the role bit; code choice, gate choice, vocabulary size, length and precision do not.

ablation	setting	acc	statistic / n
code=hadamard	code=hadamard role=True norm=True	1.0000	marginal / 512
code=random	code=random role=True norm=True	1.0000	marginal / 512
code=identity	code=identity role=True norm=True	1.0000	marginal / 512
code=gaussian	code=gaussian role=True norm=True	1.0000	marginal / 512
rho_fast	code=hadamard role=True norm=True	1.0000	marginal / 512
rho_slow	code=hadamard role=True norm=True	1.0000	marginal / 512
w_skew	code=hadamard role=True norm=True	1.0000	marginal / 512
no_norm	code=hadamard role=True norm=False	1.0000	marginal / 512
no_role_mask	code=hadamard role=False norm=True	0.6016	marginal / 512

Table 9 (part 1). Analytic ablations. The role-bit removal row is the single breaking case. A second group varies layout, encoding, gate and duplicate keys. Values for parity/gapped and duplicate keys are the five-seed entries of Table 2 (0.3438 and 0.9328); single-seed values from earlier probes are recorded only in the provenance audit (Table 10) and are not quoted as results. A learned low-rank INTERFACE of rank r >= 5 attains 0.9694 at r=4 and 1.0000 at r=16 over 3 seeds; this is a different experiment from the CODEBOOK Gram-rank collapse of Section 4.4, and the two are not compared. Finally, the adversary permutation that makes the flat distractor peak dominate gives measured accuracy 0.0000 with beta > alpha (0.94950 > 0.82605), while the codebook partition gives 1.0000; this is the concrete witness behind Lemma L1.

4.8 Numeric-provenance audit

Earlier drafts quoted several values for the same configuration. Table 10 records each drift, the single value used in this submission, its statistic, and its source. No number in this paper appears without one of these rows. **Only the five-seed column enters any claim; all superseded single-seed values are quoted for provenance only and are NEVER used as evidence.**

quantity	superseded (NOT USED)	value here	statistic / source
parity, gapped	0.2969 (1 seed); 0.2711 (other probe)	0.3438	5-seed mean / Table 2
duplicate keys	0.9316; 0.9414; 0.9328	0.9328	5-seed trained / Table 2
FFT/IIR error	2.27e-16; 2.89e-16; 2.43e-14	<= 3e-16	max over 3 probes / Table 4
chance	0.0313 and 0.25 used together	0.0313	1/V_K, this task only
attention refs	labelled chance-equivalent (withdrawn)	0.0332-0.3125	range, no chance claim / Sec 4.6
fp32 bitwise	claimed then denied	withdrawn W1	real-field only / Appendix B

Table 10. Provenance audit. The value 0.3125 in the attention range is 10/32 and is NOT 1/32; the historical four-pair task has chance 0.25 and is a different task, so 0.25 is never placed beside 0.0313. This is the single numeric convention of the paper.

4.9 Full five-seed paired statistics

Table 12 reports every five-seed configuration used in the adversarial and codebook experiments, with its mean, standard deviation and 10,000-resample paired CI. Entries are marginal accuracies under A5(ii). The paired difference between an explicit role bit and the parity format prior on the gapped layout is +0.6562 with CI [+0.6332, +0.6824] over five seeds, all positive.

mask / layout	mean	std	CI95
explicit / alternating	1.0000	0.0000	[1.0000, 1.0000]
explicit / gapped	1.0000	0.0000	[1.0000, 1.0000]
explicit / dense	1.0000	0.0000	[1.0000, 1.0000]
parity / gapped	0.3438	0.0314	[0.3176, 0.3668]
none / alternating	0.8621	0.0104	[0.8527, 0.8676]

Table 12. Five-seed statistics (duplicate identical rows in the source JSON are shown once). The per-fibre uniform value 0.0625 of Table 6 and the marginal value 0.8621 of this table are different statistics under A5; neither is a correction of the other.

4.10 Reproduction protocol

The anonymised artifact contains one reference implementation of the layer, seven minimal mechanism references used ONLY for the Section 4.6 audit, the reachability enumeration of Section 4.3, and all raw result files. All probes are CPU-only NumPy with no external model weights and complete in minutes on a single core. Seeds are listed per table: 1000+s for training, 7000+s for held-out evaluation, s in (0,..,4) for the five-seed rows; analytic witnesses use fixed seeds recorded with each table. Every number in Tables 2-12 is injected from the raw files at build time by string formatting, so a changed source file changes the PDF and a stale transcription is impossible. A single command regenerates the paper and a second command re-runs all synthetic probes. The verifier checks statement formation, witness agreement and protocol consistency. It is

NOT a mathematical proof, and all proofs in this paper are analytic.

Compute statement: total CPU time for every reported synthetic probe is under one hour; no GPU, no pretrained model, no private weight and no external dataset is used. Token counts, perplexity and generation quality are NOT measured. The 2027 call for papers and submission deadline had not been published when this version was prepared; page count, the author-side LLM-use disclosure and reciprocal-reviewer obligations must be rechecked against the official template when it appears.

4.11 Explicit falsification conditions

Each result has a stated failure mode that can be checked without re-running any training. A(b) is falsified by any exact H-hop construction with B < log2|A| on a fibre where |A| > 2^B; the identifiable place is the injection step in Appendix A.1. A(c) is falsified by a duality-preserving layer whose coefficients depend on x_t yet whose impulse response is Toeplitz, contradicting P6's induction on the lag. Result 2(i) is falsified by an order-1 state achieving exact one-hop retrieval under P2; Result 2(ii) by a single static-LTI layer exact on H >= 2 hops, or by a two-layer stack exact on H=1 under P4. P5 is falsified by an exact H-hop construction with o <= H under its d-orthogonal-successor premise. Construction C is falsified by a non-Toeplitz impulse response or an L-dependent state, both checked in Table 4. The W12 audit is falsified by reproducing the stored 0.9987 attention baseline under the Section 4.6 protocol. Each condition names the exact measurement that would have to come out differently. The Section 4.12 convergence claim is falsified by any of the five seeds missing 0.99 at step 1500 on a diagonal cell, or by any off-diagonal cell reaching it.

4.12 Convergence of the codebook-wired stack

Architectural change, not a training trick. The H6 interface of Section 2 admits the special case T_h = C_val^T C_key, the codebook's own value-to-key swap. We make this the wiring of the architecture module: it is fixed, shared by all layers and NOT a learnable parameter, so the parameter count is independent of depth. Only the external-query projection and the read-out are learned, both from random initialisation. The inter-layer unit normalisation is built from log and exp inside the graph, removing the straight-through gradient (W14). The recurrence kernel is untouched: static, Toeplitz, causal, O(1) state, no score matrix and no cache.

wiring, H=2 and two layers	acc (5 seeds)	conv. seeds	first step >= 0.99
codebook-fixed interface	1.0000 +/- 0.0000	5 / 5	150
analytic witness, zero training	1.0000 +/- 0.0000	5 / 5	0
learned interface, random init	0.2492 +/- 0.0325	0 / 5	none
learned interface, fixed swap	1.0000 +/- 0.0000	5 / 5	150
learned interface, near-swap init	1.0000 +/- 0.0000	5 / 5	150

Table 14. 1500 Adam steps, lr 0.02, batch 64, five seeds, 512 training and 512 held-out instances per seed. All arms see the same data, embedding and optimiser. The learned random interface does not converge at this budget; the codebook-fixed interface and the near-swap basin do. The swap is an algebraic object of the Hadamard dictionary and contains no training data, labels or run-time annotation.

hops / layers	1	2	3	4
1	1.0000	0.0312	0.0391	0.0363
2	0.1809	1.0000	0.0301	0.0348
3	0.1570	0.2473	1.0000	0.0363
4	0.1551	0.2414	0.3547	1.0000

Table 15. Trained layer-by-hop grid at 1500 steps, five seeds per cell. The diagonal is 1.0000 everywhere; the largest off-diagonal mean is 0.3547. The H=2 paired difference is +0.8191 with 95% CI [+0.8008, +0.8375], all five paired differences positive. Conservation, gradient and interface-identity checks pass on the final weights: Toeplitz deviation 0, FFT/IIR agreement to machine precision, future perturbations change no past output, state bytes independent of L, and finite-difference gradients within 5e-4 at every depth. These are experiments, not theorems; W13 is re-established here on the corrected architecture and the earlier number is not reused. The earlier same-run training figures in Tables 2-3 are kept as the W13 audit trail only; Tables 14-15 supersede them as the training evidence for Result 4. No language-modeling or attention-comparison claim is made.

4.13 Deep SGC-SMM fusion and provable self-evolution

The sidecar is removed. An experience is registered INTO the same Sylvester-Hadamard codebook used by the tensor state: S_t = rho S_(t-1) + sum_e C[k_e]^T C[v_e], with read-out o = q S_t. Registration is a partial isometry between two orthogonal half-spaces, hence zero-order by construction. Because Hadamard rows are exactly orthogonal, C[k] S = w_j C[V_K+v_j] when k = k_j and 0 otherwise (SE1), so retrieval accuracy is IDENTICALLY the fraction of bindings correctly registered, not a finite-sample estimate; this identity is what makes improvement provable.

arm	final acc (5 seeds)	paired diff vs fused
fused, gated	1.0000 +/- 0.0000	-
no consolidation	0.0000 +/- 0.0000	+1.0000
no gate, 30% corrupt	0.6438 +/- 0.0781	+0.3563
broken alignment	0.0250 +/- 0.0364	+0.9750
fast decay	0.1313 +/- 0.0234	+0.8688

Table 16. Seven arms, 320 rounds, five seeds, a one-sided paired-CI gate (alpha = 0.05). The fused arm reaches 1.0000 on all five seeds with a monotone curve on all five; removing consolidation gives 0, and removing the gate under 30 percent corruption gives 0.6438. SE2's coupon-collector bound predicts E[T_cover] = 129.87 rounds against a five-seed mean of 130.00 (relative gap 0.10 percent, 95% t-CI [82.72, 177.28] rounds, n = 5). The mean agrees with the prediction, but the seed-to-seed spread is large: the small relative gap is a coincidence of the MEAN and is NOT a precision claim. The premise was fixed to i.i.d. arrival BEFORE measurement, not fitted after. SE4 caps the information entering the state at T w log2(V_K) bits, independent of context length. All conservation checks pass on the FINAL post-evolution state (Toeplitz deviation 0, FFT/IIR to machine precision, causality, state independent of L). Full trajectories and the gate x corruption grid are Tables 18-19 in Appendix A.7-A.11. These are theorems and measurements on an exactly-defined finite synthetic process, NOT language modelling or general self-improvement.

5 What This Does NOT Establish

· **No parity claim.** W12 withdraws it; the same-run audit did not reproduce the stored baseline. No equivalence, superiority or inferiority is claimed.

· **No architecture comparison.** Section 4.6 supplies same-condition toy mechanism references for a reproduction audit, not full reproductions of GLA, DeltaNet, Mamba-2 or Transformer, and not a performance ranking. Earlier O1-O4 lacked 5-seed paired CIs; the Section 4.6 results have them.

· **No language modeling.** There is no >=10M-token equal-compute run, no val perplexity, no generation readability and no measured throughput comparison.

· **Not a new paradigm.** P4 states the rho=1 single-pole isomorphism with linear attention. The contribution is a tight JOINT characterization under static kernels.

· **Role necessity is conditional.** With no role bit the tight alternating layout gives 0.8621 while a gapped layout is higher because zero fillers suppress the spurious peak (L1). Under the codebook partition neither a tag nor a position rule is needed; the alphabet convention remains an encoding-level inductive bias. Exactly one bit per token is a sufficient carrier, not an information-theoretically minimal one.

· **Overshoot is not an architectural law.** Theorem A(f) attributes H_L > H failure to last-layer read-out, not to the static kernel family.

· **Self-evolution is bounded.** Section 4.13 proves real, monotone improvement for a finite consolidation process, but SE4 caps the information entering the state at T w log2(V_K) bits, independent of context length and bank capacity. Long context buys inference-time retrieval, not a faster learning rate. No language-model, perplexity or general-capability improvement is claimed.

6 Related Work, Placed Precisely

Table 1 gives the exact increment per known result; we do not repeat it here. Zoology [1], Repeat After Me [2] and Illusion of State [3] established that fixed-state models struggle with associative recall. S5 [4] gave the LTI duality and the requirement of time-invariant coefficients; Mamba [5]

observed that selectivity breaks the convolution view; linear attention [6] and its gated and delta-rule variants [7,8] use outer-product states. We do not claim to be the first to observe that static convolutions struggle with recall, nor that LTI systems require time-invariance. The increment is the exact algebraic bound B >= log2 |A(v)|, its per-fibre accuracy statement, the tight binary-chain witness, and the joint layer/state characterization under a strictly static kernel.

6.1 Relation to MQAR and associative-recall protocols

Zoology, MQAR and induction-head tasks measure recall under a large vocabulary, multiple queries and non-adjacent bindings, with performance summarised over a distribution of sequences. The task studied here is deliberately narrower: n_pairs = 4, V_K = 32, one external query, adjacent binding emission and a single last-layer read-out. It was chosen because it admits an EXACT per-fibre answer set A(v) that can be enumerated, which is what makes the bound tight and falsifiable. It is not an MQAR result and is not reported as one. In particular, the 0.9512 five-seed trained accuracy on H=2 (Table 3) is not comparable to an MQAR score: the vocabularies, query counts, sequence lengths and answer distributions differ. What transfers is the information bound A(b), which is distribution-free and therefore applies to MQAR as well: any flat projection that leaves |A(v)| compatible answers requires at least log2|A(v)| input-side bits, regardless of vocabulary size. Establishing the same tightness on a standard MQAR protocol would require enumerating A(v) there, which we did not do and do not claim.

7 Limitations and a Falsifiable Prediction

Theorem A assumes distinct keys, adjacent binding emission, an external query, last-layer read-out and a deterministic decoder; multi-query chains and stop/select mechanisms are outside Result 2. Proposition P5's state-order bound is conditional on c mutually orthogonal candidate successors per hop; when the branching factor is 2 it degenerates to 2^(H+1), which we state rather than hide. Finite-precision results cover L <= 128 for the analytic construction; the long-decay frontier at L=2049 is a separate measurement. Theorem A(b) is distribution-free and depth-free: no deterministic decoder supplied with B < log2 |A(v)| input-side bits can be exact on all compatible instances. Any construction attaining exact H-hop retrieval with fewer bits falsifies the bound at a precisely identifiable place. We did not run a >=10M-token equal-compute language-model experiment, and we do not present this as an architecture paper. A learnable codebook, a converged strong baseline and a fair throughput comparison remain open. That absence is a target rather than a request for leniency: the consolidation rate bound (SE4, Appendix A.10) gives a necessary bound on how fast any such run can inject information into a constant-size state, T w log2(V_K) bits in T rounds, independent of context length. Any run exceeding this rate without enlarging the state falsifies the bound at an identifiable place; we invite the community to falsify it under a published compute budget. The tight order/depth bounds of Result 2 are likewise falsifiable by a single order-1 state or a single layer exact on H >= 2, conditions listed in Section 4.11.

8 Conclusion

Exact content addressing needs an order-2 tensor state, and nothing smaller works. We proved the matching lower bound and the attaining construction (Result 2), showed that the same static-LTI layer stack needs H layers for H hops under Result 2(ii)'s premises and with O(1) state in L, and verified both predictions by a zero-training 5 x 5 grid and a five-seed 1500-step run. The supporting input-side bound (B >= log2 |A(v)|, Theorem A) quantifies the bit budget this construction presupposes; on the content-chain task it is loose (Section 4.3), and what transfers is the distribution-free inequality A(b). The inter-layer map is fixed to the codebook's own value-to-key swap, contains no training data and leaves the recurrence kernel Toeplitz, causal and static. The earlier attention-parity claim is withdrawn because it did not reproduce (W12). Deep fusion with zero-order consolidation (Section 4.13, Appendix A.7-A.11) shows monotone self-improvement on an exactly-defined finite process while remaining inside the same information bound. The practical rule is to budget log2|A(v)| bits at the input, then choose H layers under Result 2(ii)'s premises and an order-2 state, or relax exactness. This is a tight boundary-characterization paper with a constructive counterexample, not a claim of language-model parity, equivalence, superiority or a new SOTA sequence model.

Broader Impact

Generative AI tools were used for implementation assistance and language editing; all research questions, proofs and conclusions were specified and verified by the human author. This disclosure is subject to the ICML 2027 author instructions, which are not yet published.

This is a theoretical and synthetic-experimental study of information limits in fixed-state sequence models. It contains no language-model training, no human-subject data, no deployed system and no generative capability. The immediate practical consequence is a design rule: an input-side information budget must be accounted for before attributing a retrieval failure to model capacity. A possible misuse is to quote the synthetic exactness figures as evidence of general language-model parity, or to treat the withdrawn parity claim as a performance result; both readings are excluded explicitly in Sections 5 and 8. The work does not introduce a new mechanism for surveillance, data collection or large-scale deployment. Its environmental footprint is limited to CPU-only synthetic probes.

References

[1] Arora, S. et al. Zoology: Measuring and Improving Recall in Efficient Language Models. ICLR 2024. arXiv:2312.04927.

[2] Jelassi, S. et al. Repeat After Me: Transformers are Better than State Space Models at Copying. ICML 2024. arXiv:2402.01032.

[3] Merrill, W. et al. The Illusion of State in State-Space Models. ICML 2024. arXiv:2404.08819.

[4] Smith, J. T. H., Warrington, A., and Linderman, S. W. Simplified State Space Layers for Sequence Modeling. arXiv:2208.04933, 2022.

[5] Gu, A. and Dao, T. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. COLM 2024. arXiv:2312.00752.

[6] Katharopoulos, A. et al. Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention. ICML 2020. arXiv:2006.16236.

[7] Yang, S. et al. Gated Linear Attention Transformers with Hardware-Efficient Training. ICML 2024. arXiv:2312.06635.

[8] Yang, S. et al. Parallelizing Linear Transformers with the Delta Rule over Sequence Length. NeurIPS 2024. arXiv:2406.06484.

[9] Dao, T. and Gu, A. Transformers are SSMs: Generalized Models and Efficient Algorithms Through Structured State Space Duality. ICML 2024. arXiv:2405.21060.

[10] Poli, M. et al. Hyena Hierarchy: Towards Larger Convolutional Language Models. ICML 2023. arXiv:2302.10866.

[11] Romero, D. W. et al. CKConv: Continuous Kernel Convolution For Sequential Data. ICLR 2022. arXiv:2102.02611.

[12] Tay, Y. et al. Long Range Arena: A Benchmark for Efficient Transformers. ICLR 2021. arXiv:2011.04006.

[13] Vaswani, A. et al. Attention Is All You Need. NeurIPS 2017. arXiv:1706.03762.

[14] Olsson, C. et al. In-context Learning and Induction Heads. Transformer Circuits Thread, 2022.